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Roll No

MI-7002-CBGS

B.E. VII Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Strata Control

Time : Three Hours

Maximum Marks : 70

Note : i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) A borehole log of 2.6 m in length, recovery of rock cores in cm are 15, 16, 15, 12, 6, 13, 9, 10, 12, 15, 8, 5, 19. Then find out the Core Recovery and RQD value of rock in percentage. 2
b) Design support system for Galleries, Junctions and Depillaring workings of a Bord and Pillar mine. Condition of mine is as following: 12
RMR = 52, Gallery span = 4.2 m, Height = 3m, Density = 2.1 t/m³, Jn = 9, Jr = 1.5(Assume the required data if any).
2. a) Describe any one of the powered support and its principles of operation in details. 7
b) Describe a hydraulic prop and its working principle. Compare it with a friction prop. 7
3. a) Explain the theories of "Roof bolting". 7
b) Write a note on "Roof stitching" and "Hydraulic fluids". 7
4. a) What are the advantages of stowing? Describe the process of hydraulic stowing with its advantages and disadvantages. Describe the process of hydraulic stowing in a Bord and Pillar mine (with sketch). 2+2+7

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- b) A goaf void of 250 m^3 is filled in 3 hours by hydraulic sand stowing method. Density of the sand is 2.6 tonne/m^3 . If the filling factor of goaf void is 0.9 and sand to water ratio in the stowing mixture is 1.0 tonne to 1.1 m^3 , then calculate the stowing rate in m^3/h . 3
5. a) Write a note on the behaviour of rock pressure during narrow and wide excavation. 7
- b) Explain the theories of "Ground movement". 7
6. a) Explain in detail the development of subsidence caused due to a longwall panel with a neat sketch indicating caving, fracturing and subsidence in the strata. 7
- b) Define/Explain in respect of subsidence 7
- i) Strain
- ii) Tilt
- iii) Point of Inflexion
- iv) Horizontal displacement
7. a) At a depth of 750 m, a 10-m diameter circular tunnel is driven in rock having a unit weight of 26 kN/m^3 and uniaxial compressive and tensile strengths of 80.0 MPa and 30.0 MPa respectively. Will the strength of the rock on the tunnel boundary be exceeded if:
- i) $k = 0.3$
- ii) $k = 2.0$ 7
- b) Discuss various factors affecting the mine subsidence. 7
8. Write short notes on (any four) 14
- a) Strata Control and Monitoring plan
- b) Support plan
- c) Withdrawal of support
- d) Rate of stowing
- e) Pipe jam in hydraulic stowing

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